

Kaa: This is platform-centric middleware. It manages an unlimited number of connected devices with cross-device interoperability. It performs real-time device monitoring, remote device provisioning and configuration, as well as collection and analysis of sensor data. It supports on-premise, public cloud and hybrid models of deployment.

IoTSyS: IoTSyS is platform-centric and industry-specific middleware. It uses IPv6 for non-IP IoT devices and systems. Used in smart city and smart grid projects, IoTSyS provides interoperable Web technologies for sensors and actuator networks.

DeviceHive: This cloud-agnostic, microservices based, platform-centric middleware is used for device connectivity and management. It has the ability to connect to any device via MQTT, REST or WebSockets. It supports Big Data solutions like Elasticsearch, Apache Spark, Cassandra and Kafka for real-time and batch processing.

Zetta: This application-centric middleware is a service-oriented open source IoT platform built on Node.js that combines REST API, WebSockets and reactive programming. It can run on cloud platforms like Heroku to create geo-distributed networks and stream data into machine analytics software like Splunk.

OpenIoT: This is an application-centric open source middleware for pulling information from sensors. It acts as sensing-as-a-service for deploying and providing services in the cloud environment.

Other popular open source middleware platforms include ThingsBoard, NATS, DSA, MainFlux, OpenIoT, Sitewhere, OpenRemote and EclipseIoT.

Open source security for the IoT ecosystem

Security is the key parameter that needs to be addressed while setting up the IoT system. Security helps to automatically discover IoT devices and analyse their

network traffic. IoT security features must be added at the start of the device development process, and included in its hardware and firmware.

Popular open source security tools include Princeton IoT Inspector, Open Web Application Security Project, Mainflux, etc.

Open source databases and analytics

The entire implementation of IoT moves around the idea of data collection and insertion through sensors, and sending control back to the devices. The selection of the right database in various stages of IoT implementation for a given IoT use case is important and is dependent on multiple factors like the extent of enterprise adoption, the nature of the business, etc.

The following are a few top open source databases available for IoT based applications.

MongoDB: This is a powerful, flexible, document-oriented and scalable NoSQL database. It supports features like range queries, sorting, aggregations and geospatial indexes. It also supports JSON to store and transmit information. JSON, being a standard protocol, is a great advantage for both the Web and the database. MongoDB supports a rich query language for read and write operations (CRUD) as well as data aggregation, text search and geospatial queries.

MySQL: This open source relational database management system brings data consistency, scalability, high performance, availability and flexibility to the IoT solution by efficient collection of data from IoT devices. It helps in data transformation through annotation and aggregation, for better analysis.

Redis: This in-memory open source database is a popular choice for IoT solutions. It is widely used for real-time analytics, messaging, caching, and many other use cases. It also helps in edge computing, which involves deep learning, image recognition and other innovative computing requirements.

Cassandra: This is a highly scalable and distributed open source database for managing voluminous amounts of structured data across many commodity servers. It provides availability, linear scale performance, simplicity and easy distribution of data across multiple database servers. It supports strong data consistency across distributed architectures.

Hadoop: This is an open source software platform for distributed storage and distributed processing of very large data sets on computer clusters built from commodity hardware. It helps in driving analytics from all IoT data. It easily ingests data from multiple data sources, and supports both batch as well as real-time data ingest from sensors using tools such as Apache Kafka and Apache Flume. It handles multiple IoT data types, structures, and schemas. It also supports real-time processing and applications on streaming data. Hadoop is a flexible, scalable and secure database.

Other popular open source IoT databases are CrateDB, GridDB, InfluxDB, BigChainDB, etc.

Open source connectivity protocols

IoT protocols are a crucial part of the IoT technology stack. Without them, hardware would be rendered useless as these protocols enable it to exchange data in a structured and meaningful way. IoT primarily exploits standard protocols and networking technologies.

The top five open source connectivity protocols in use today are briefly described below.

CoAP (Constrained Application Protocol): CoAP is an Internet application protocol for constrained devices. It is designed for IoT systems based on HTTP protocols. CoAP makes use of the UDP protocol RESTful architecture. It is used in mobiles and in social network based applications.

MQTT Protocol: This messaging protocol is used for remote monitoring in IoT. Its primary task is to acquire data from many devices and transport it to